

THE INVENTOR

Buddy R. Paul

Distinguished Engineer 6 Chief Systems Architect
Industry · Aerospace · NASA · Clean Energy



Buddy R. Paul, VIVIFY's Chief Systems Architect, is a career design engineer whose work spans more than four decades and some of the most demanding programs in modern engineering, including NASA's Space Shuttle — where he engineered the fix for a cryogenic fuel leak that threatened the program — the THAAD missile-defense system, Airbus aircraft systems, and Lockheed Martin, where he was named **Inventor of the Year in 2003**. His experience runs across aerospace, defense, energy, electronics, hydraulics, pneumatics, and nuclear systems, and includes roughly **250 patents and patent-related technologies**. The pages that follow document a verified core of that record through the USPTO.

40yrs

OF ENGINEERING &
INVENTION

-250

PATENTS & PATENT-
RELATED TECHNOLOGIES

9

DOCUMENTED ISSUED U.S.
PATENTS

4

LANDMARK PROGRAMS:
SHUTTLE, THAAD, AIRBUS

"My passion has always been design. I like identifying the need for something and then coming up with the answer for that need."

BUDDY R. PAUL

A CAREER OF INVENTION

One discipline, applied to three hard problems.

■ 01 STRUCTURAL ENGINEERING · 1980S · CLARY CORPORATION

Building the foundations

Paul's first granted patents were mechanical: a pair of 1986 design patents for a truss connector plate, assigned to Clary Corporation. Early proof of a mind drawn to load, structure, and how parts hold together under stress — the same instincts that would define everything he built next.

Representative - US D286,855 • US D287,223

■ 02 AEROSPACE, DEFENSE & NASA · 1990S-2000S

The engineer who fixed the cryogenic leak

On NASA's Space Shuttle program, Paul engineered the solution to a persistent cryogenic liquid-hydrogen leak — the kind of failure that grounds a fleet. The same precision-mechanism thinking runs through his work on the Terminal High Altitude Area Defense (THAAD) missile system and Airbus aircraft systems, and spans electronics, hydraulics, pneumatics, and nuclear applications. His granted patents from this era are missile-launch systems built at Lockheed Martin: a passive dynamically disconnecting arm, a rocket-launcher docking system, a passively-actuated snubber, and a canister closure that releases on a pressure pulse that was engineered to perform once, flawlessly, in the split second of a launch. In 2003, Lockheed Martin named him its **Inventor of the Year**.

Representative - US 6,679,154 • US 7,070,434 • US 7,624,669 • US 7,685,920

A complete energy architecture

Paul then turned the same discipline on the energy problem. Across more than thirty patent filings he built out a full clean-energy architecture — hydrogen generation and electrolysis, emissions capture, turbines, and combustion — and secured three issued U.S. patents at its core. That body of work is the engineering foundation beneath VIVIFY's HOG™, CAT™, and Flying Pig™ platforms.

Representative - US 9,038,595 B2 • US 9,624,801 B2 • US 9,664,139 B2

THE CLEAN-ENERGY CORE

Three issued patents power VIVIFY's platforms.

US 9,038,595 B2

Carbon Oxygen Hydrogen Motor

The hydrogen-combustion engine at the heart of the HOG™ generator platform.

HOG™ • Hydrogen Energy

US 9,624,801 B2

Exhaust Mitigating System & Power

Removes contaminants from exhaust gas and generates electrical power: the basis of CAT™.

CAT™ • Emissions Control

US 9,664,139 B2

Captive Oxygen Fuel Reactor

The reaction core that drives on-demand, water-based hydrogen power generation.

Reaction Core

APPENDIX A

Issued United States patents

PATENT NO.	INVENTION	FIELD	ASSIGNEE	ISSUED
US D286,855	Connector plate for trusses	Structural	Clary Corp.	1986
US D287,223	Connector plate for trusses	Structural	Clary Corp.	1986
US 6,679,154	Passive dynamically disconnecting arm	Aerospace & Defense	Lockheed Martin	2004
US 7,070,434	Rocket-launcher docking system	Aerospace & Defense	Lockheed Martin	2006
US 7,624,669	Apparatus comprising a passively-actuated snubber	Aerospace & Defense	Lockheed Martin	2009
US 7,685,920	Canister closure with pressure-pulse release	Aerospace & Defense	Lockheed Martin	2010
US 9,038,595	Carbon Oxygen Hydrogen Motor	Clean Energy		2015
US 9,624,801	Exhaust Mitigating System & Power	Clean Energy		2017
US 9,664,139	Captive Oxygen Fuel Reactor	Clean Energy		2017

APPENDIX B

Clean-energy portfolio

The full body of clean-energy filings — hydrogen, emissions, turbines, and combustion — grouped by domain. Granted patents are marked. Serial numbers prefixed 61/ and 62/ are provisional filings.

INVENTION	APPLICATION	PUBLICATION/ GRANT
HYDROGEN 6- ELECTROLYSIS		
Electrolysis Apparatus and Related Devices and Methods	12/909,813	US 2011/0100328 A1
Hydrogen Oxygen Generator	61/256,129	US 2011/0100328 A1
Hydrogen Oxygen Combustion Alternator	61/258,102	US 2011/0100328 A1
Hydro Electric River Oxygenator	61/495,257	US 2011/0100328 A1
Alternating Direct Current Electrolyzer	61/548,334	US 2011/0100328 A1
Hydrogen Generator Battery	61/559,550	US 2011/0100328 A1
Solar Wind Electrolyzer Electrical Turbine	62/050,486	US 2011/0100328 A1
Solar Wind Electrolyzer Electrical Turbine	62/202,348	US 2011/0100328 A1
COMBUSTION, MOTORS 6- FUEL		
Carbon Oxygen Hydrogen Motor	13/444,594	US 9,038,595 B2 Granted

INVENTION	APPLICATION	PUBLICATION/	GRANT
Captive Oxygen Fuel Reactor	15/205,721	US 9,664,139 B2	Granted
Captive Oxygen Fuel Reactor	62/190,052	US 9,664,139 B2	Granted
Captive Oxygen Fuel Reactor	62/022,071	US 2017/0009702	A1
Carbon Oxygen Hydrogen Motor	61/474,147	US 2012/0255518	A1
New Internal Combustion Engine	61/258,103	US 2011/0100328	A1
Carbon Compound Rod	61/320,380	US 2011/0100328	A1
Spark Oxygen Hydrogen Injection Plug	61/453,382	US 2011/0100328	A1
Turbine Jet	61/542,646	US 2011/0100328	A1
CLEAN AIR & EMISSIONS			
Mitigating System to Remove Contaminants from Exhaust Gas & Generate Power	14/719,540	US 9,624,801 B2	Granted
Clean Air Turbine	61/440,078	US 9,624,801 B2	Granted
Clean Air Turbine	61/598,376	US 9,624,801 B2	Granted
Artificial Gravity Enhance Separator	61/497,365	US 9,624,801 B2	Granted
Particle Acquisition Tower	61/497,618	US 9,624,801 B2	Granted
Cryogenic Air Pump	61/497,623	US 9,624,801 B2	Granted
Natural Ecological Way Mitigating System	61/497,361	US 2011/0100328	A1
Recycle Incinerator Generator	61/512,815	US 2011/0100328	A1
TURBINES, POWER & SYSTEMS			
Heat Exchange Turbine Generator	61/527,970	US 2011/0100328	A1
AC-DC Voltage and Current Doublers	61/321,165	US 2011/0100328	A1
Hygienic Induction Cauldron	61/505,928	US 2011/0100328	A1
WATER & PROCESS			
Digital Electronic Water Conditioner	61/485,685	US 2011/0100328	A1
Method of Reducing Salt and Sugar Content in Food	61/536,337	US 2011/0100328	A1

Sources: United States Patent and Trademark Office (USPTO); Justia Patents inventor record for Buddy R. Paul; VIVIFY Technology patent list.

VIVIFY · BRINGING TECHNOLOGY TO LIFE

**Space Shuttle.
THAAD. Airbus. Now
VIVIFY.**